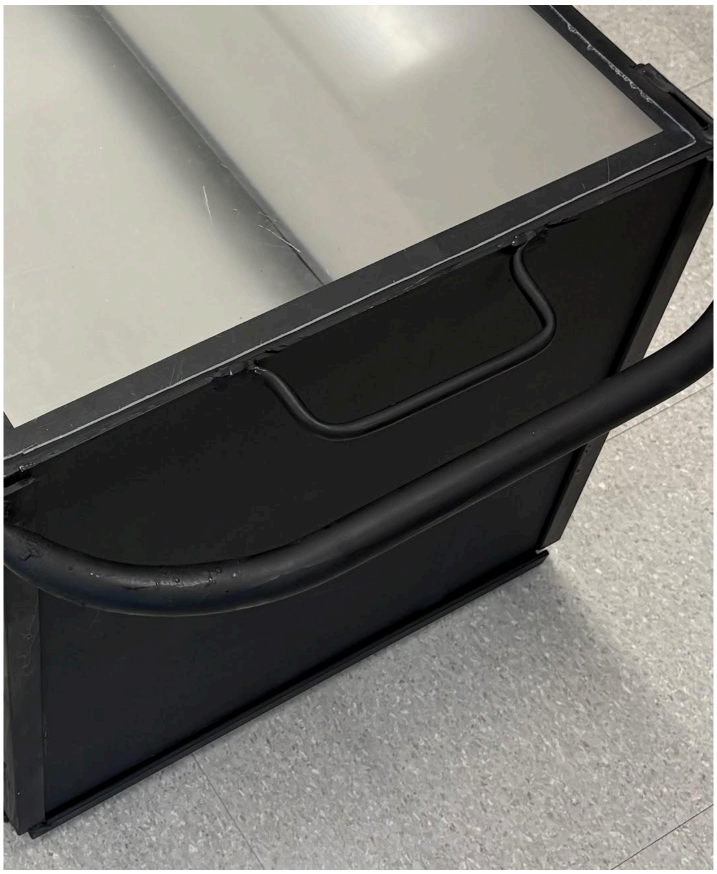
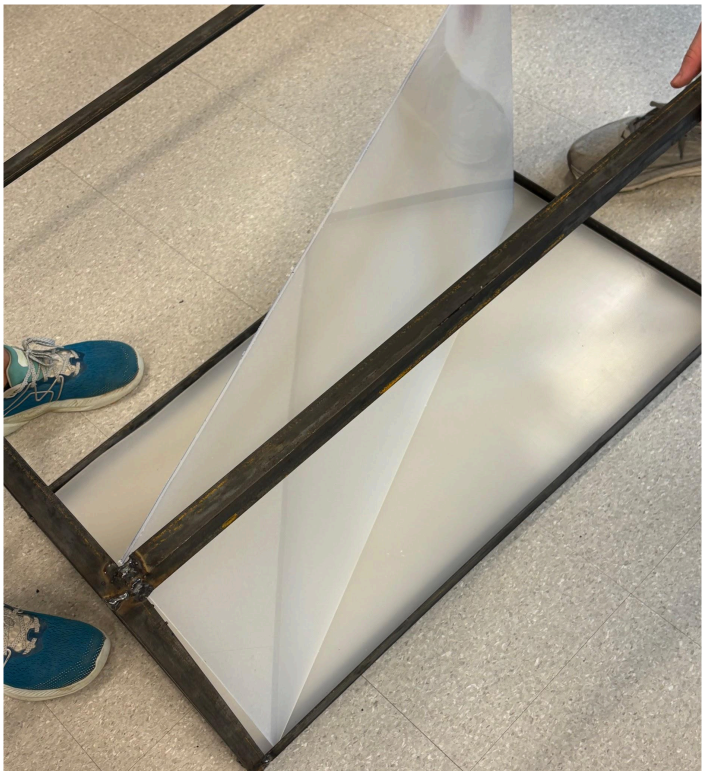
Komodo Dragon Element L

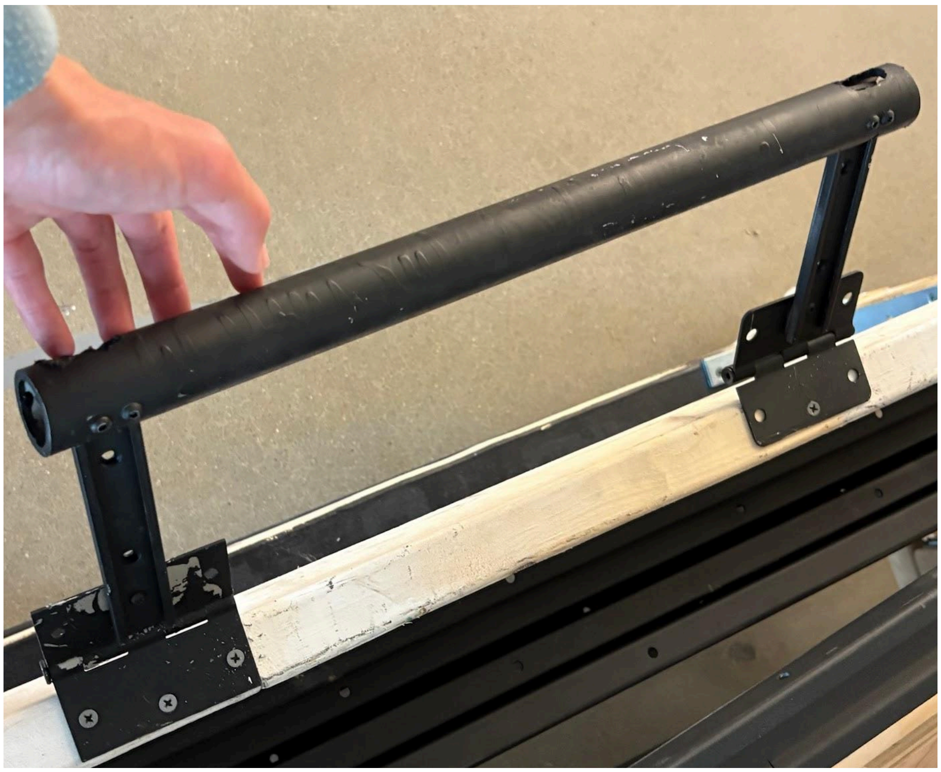
Though we did complete a final prototype, there were many areas of improvement that could take our project to the next level.



For one thing, we could have also researched a way to help the guillotine style door slide into place better. The current guillotine style door can get stuck and become hard to pull out of the slot. It can also be tricky to slide it back in if it gets caught. In the future, we maybe could have added some sort of slick lube on the sides of the door and inside the slot. Or we could have welded the slot down so the door had more space to slide in. This would probably also remove the problem of the guillotine door getting stuck.



Another thing that would be a good improvement in the future would be to maybe use a little bit thicker plastic. The only reason that we were able to get the thin plastic was because of budgeting problems. We still did run tests and the plastic that we have now should work just fine. However, when the zoo experts came down to examine our project, this was their only concern. They didn’t seem too concerned about the floor or ceiling, just the walls. Maybe in the future, we would have bought the thicker ¼” plastic and used some of our own money to pay for the extra plastic. Another thing we could have done to add support to the walls would have been to add an extra steel brace going vertically down the center of the sides to prevent fluctuation in the walls and to make our solution more viable. Another problem with the thickness of the plastic walls was that it was the only section of the plastic that had a chance to crack while we were drilling holes. The very first time we were drilling and went too fast, a crack formed near the hole in the thinner plastic. This never happened in the thicker plastics. This is another reason why in the future, maybe thicker plastic should be used for all sides and not just the floor.



Another improvement that we could have made to our project would be to upgrade the handles. Though the handles do their job as they were able to lift this approximately 300 pound cart off of the ground fairly easily, there are many flaws with the handles. For example, they don’t look as presentable as the rest of the rest of the box. The spray paint didn’t get all over the hinges so there are still some gray spots there and there is a weird layering of paint on the handles that makes holding them feel a little bit awkward. Also, because we used rustoleum spray paint to try to make the handles look better and flow with the appearance of the box itself, it makes the handles feel a little sticky when you are holding it. In the future, we could have researched some paints that didn't feel sticky to hold after they are dry, if there are any. Another problem with the handles was that even when they were tightened in with nuts and bolts, they are still able to lateral move back and forth which can be concerning. We know that the handles can hold the weight, so we aren't concerned about anyone's safety. In the future, maybe we should have been more precise with our measurements so that the hinges connected to the handles had little to no wiggle room unlike some of the handles that we have now. These are just some ways to make improvements on the frame and handles of the box. There are probably other ways to improve the connection system and the medical testing door, but those were the tasks of the other groups.